ADR-001 · Embeddings Sourced From the Hugging Face Inference Router, Not the Local Custom LLM Endpoint

Field	Detail
Decision	Use langchain_huggingface.HuggingFaceEndpointEmbeddings (HF Inference Providers router) for all embedding calls in 3_rag_v1.py, rather than routing embeddings through the project's local CUSTOM_API_BASE chat server
Date	2026-07-08
Context	The local server only implements /v1/chat/completions, /v1/completions, and /health (confirmed via its /openapi.json, see Bugs.md BUG-001) — it has no embeddings route under any configuration. An embeddings client was needed that didn't depend on standing up a second local server or model.
Options Considered	Add an embeddings endpoint to the local TGI-style server (out of scope, external dependency) · fall back to OpenAIEmbeddings against the real OpenAI API (works, but reintroduces the cloud cost/key dependency the local-LLM setup was meant to avoid) · HuggingFaceEndpointEmbeddings via the HF Inference router (free tier, HF_TOKEN already available, confirmed working via a direct curl test)
Chosen Solution	HuggingFaceEndpointEmbeddings(model="sentence-transformers/all-MiniLM-L6-v2", provider="hf-inference", huggingfacehub_api_token=HF_TOKEN), model name overridable via HF_EMBEDDING_MODEL_NAME
Rationale	Reuses credentials/infrastructure already present in .env rather than adding a new provider; keeps the chat LLM fully local while sourcing the one capability (embeddings) the local server can't provide from a free, already-available router.
Impact	3_rag_v1.py embeddings block rewritten; langchain-huggingface added as a dependency, pinned <1.0.0 for langchain-core compatibility. Chat generation (llm_setup.py) is unaffected. Note: 3_rag_v4.py still uses OpenAIEmbeddings and has not been migrated under this decision — see Architecture.md module breakdown for 3_rag_v4.py.

ADR-002 · Project .venv Built on Python 3.12, Not 3.14

Field	Detail
Decision	Create the project virtual environment with Python **3.12** (`py -3.12 -m venv myenv`) rather than the newer 3.14
Date	2026-07-08
Context	`python3` / `python3.11` were not runnable at all — no real Python was on `PATH` (Windows Store alias redirect) and no 3.11 runtime was registered with the `py` launcher (`py -3.11 --version` → "No runtime installed that matches 3.11"). `py --list` / `py -0` showed only two real installed runtimes: 3.14.3 and 3.12.10.
Options Considered	Python 3.14 (newest installed, but very recently released — higher risk of dependency wheels/compatibility gaps with the LangChain/LangSmith stack) · Python 3.12 (installed, one minor version behind, broader ecosystem/wheel support at time of setup) · install Python 3.11 via `py install 3.11` (matches what was originally attempted, but adds an extra install step for no clear benefit over 3.12)
Chosen Solution	`py -3.12 -m venv myenv`
Rationale	3.12 offered the best balance of "already installed" and "broadly compatible with LangChain/LangSmith packages" without gambling on a same-week-class Python release (3.14) having full wheel support from every dependency.
Impact	`.venv` in the project root targets Python 3.12.10. No code depends on 3.12-specific syntax; revisiting this choice later (e.g. moving to 3.14) should be low-risk if a concrete reason arises.

ADR-003 · Remaining OpenAI-Dependent Scripts Migrated Onto the Local TGI Proxy + HF Router Stack

Field	Detail
Decision	Migrate 3_rag_v4.py (embeddings + chat), 4_agent.py (chat), and 5_langgraph.py (chat) off the OpenAI API entirely, onto the same llm_setup.llm (local TGI Proxy) + HuggingFaceEndpointEmbeddings (HF Inference router) pattern already used by 3_rag_v1–v3 (ADR-001)
Date	2026-07-09
Context	The user's OpenAI API key stopped working. .env was intentionally trimmed to drop OPENAI_API_KEY (along with GROQ_* and several other unused vars) as part of a cleanup pass, which meant 3_rag_v4.py (OpenAIEmbeddings + bare ChatOpenAI("gpt-4o-mini")), 4_agent.py (bare ChatOpenAI()), and 5_langgraph.py (bare ChatOpenAI("gpt-4o-mini")) would fail outright — they were the only scripts still depending on OpenAI credentials.
Options Considered	Restore OPENAI_API_KEY and keep troubleshooting the account/billing issue (rejected — key is confirmed non-functional and out of this project's control) · leave the three scripts broken/unmaintained (rejected — they're active teaching demos) · migrate them onto the already-proven local-LLM + HF-embeddings stack (chosen)
Chosen Solution	3_rag_v4.py: OpenAIEmbeddings → HuggingFaceEndpointEmbeddings (default sentence-transformers/all-MiniLM-L6-v2), ChatOpenAI → llm from llm_setup. 4_agent.py: ChatOpenAI() → llm from llm_setup. 5_langgraph.py: ChatOpenAI(...) → llm from llm_setup; with_structured_output replaced with a prompt + PydanticOutputParser approach (forced by BUG-002, discovered while testing this migration).
Rationale	Keeps the whole project on one credential set (CUSTOM_API_BASE/CUSTOM_API_KEY/HF_TOKEN) instead of two, removes a dependency on a currently-broken external account, and reuses a pattern already validated end-to-end in 3_rag_v1.py. Verified each migrated script actually runs against the local stack before considering this done (see Status.md 2026-07-09 entry).

Impact	3_rag_v4.py, 4_agent.py, 5_langgraph.py rewritten; dead OpenAIEmbeddings/ChatOpenAI imports also removed from 3_rag_v2.py/3_rag_v3.py. OPENAI_API_KEY is no longer required anywhere in this project. Supersedes the "not yet migrated" note this project's Architecture.md previously carried for 3_rag_v4.py. Related: BUG-002, BUG-003.

ADR-004 · Tracking Docs Moved Into docs/ and No Longer Git-Ignored

Field	Detail
Decision	Move Status.md, Architecture.md, Decisions.md, Research.md, Bugs.md (plus the course transcript and the tracking guide) into a docs/ folder, and remove the .gitignore rule that had excluded the five tracking files from version control
Date	2026-07-09
Context	The tracking system was originally set up mirroring RAG-work's convention of keeping these five files git-ignored (local-only, never committed). The user restructured .gitignore (also dropping an unrelated Graphify/Claude-specific section) and relocated the docs into docs/ without re-adding that exclusion.
Options Considered	Keep the tracking docs git-ignored/local-only, matching RAG-work (rejected — no longer what .gitignore reflects) · commit docs/ as part of the repo so history/decisions/bugs travel with the code and are visible in PRs (chosen, matches the user's actual .gitignore edit)
Chosen Solution	docs/ is a normal, tracked folder; no gitignore rule excludes it or the five tracking files
Rationale	Reflects the change the user already made rather than fighting it; since .env (the only place secrets live) remains git-ignored, committing the docs doesn't create a credential-leak risk — redacted references like CUSTOM_API_BASE/HF_TOKEN (env var names, not values) are the only credential-adjacent content in these files.
Impact	Future commits will include docs/*.md changes; anyone cloning the repo gets the full Status/Architecture/Decisions/Research/Bugs history. No code impact.